Appendix: Bluetooth LE

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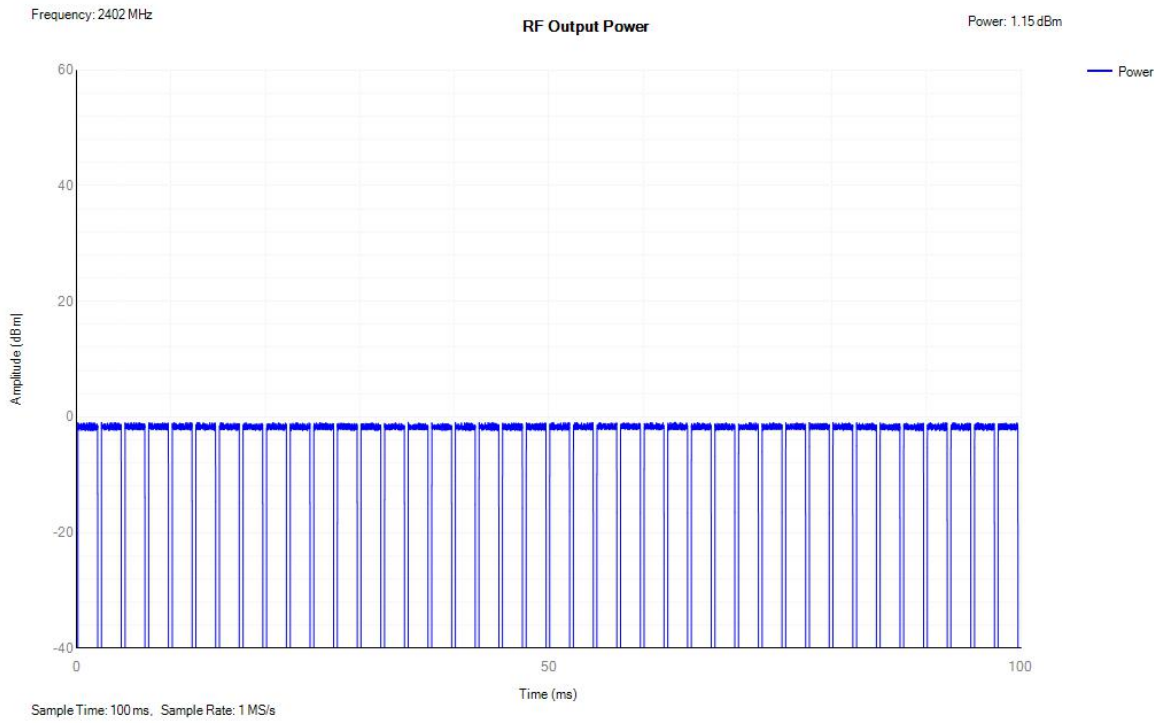
RF Output Power

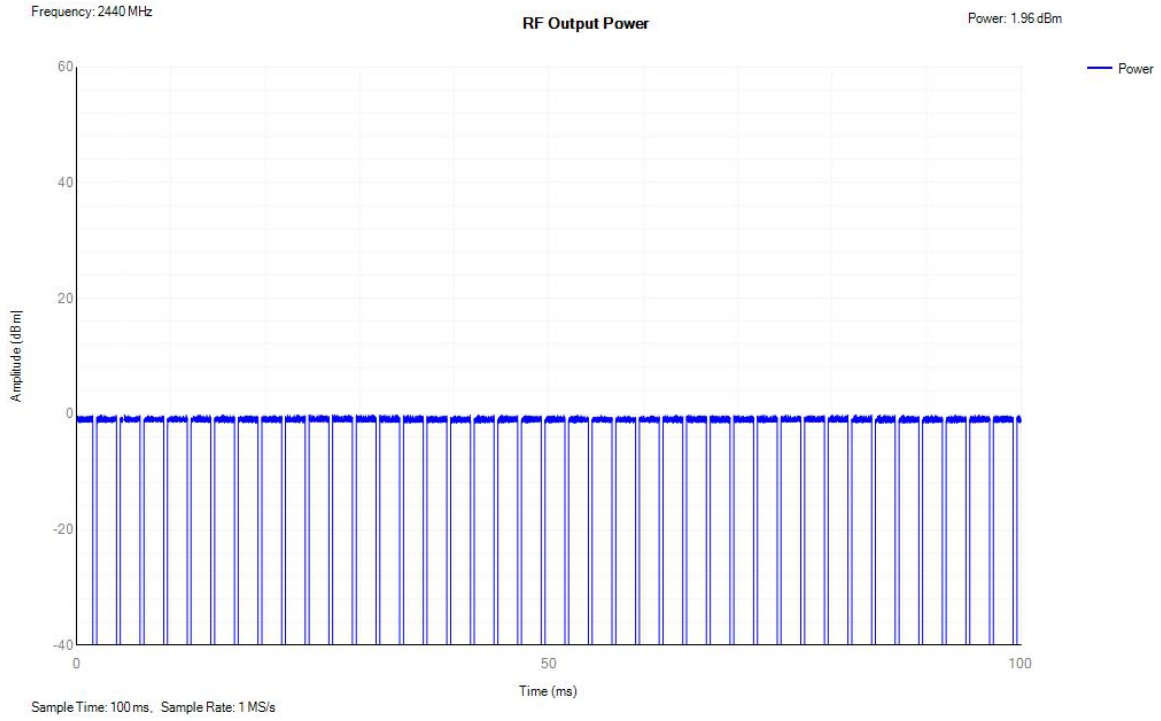
Condition	Mode	Frequency (MHz)	Antenna	Max Burst RMS Power (dBm)	Burst Number	Max EIRP (dBm)	Limit (dBm)	Verdict
NVNT	BLE 1M	2402	Ant1	-1.75	40	1.15	20	Pass
NVNT	BLE 1M	2440	Ant1	-0.94	41	1.96	20	Pass
NVNT	BLE 1M	2480	Ant1	-0.84	41	2.06	20	Pass
NVLT	BLE 1M	2402	Ant1	-1.68	41	1.22	20	Pass
NVLT	BLE 1M	2440	Ant1	-0.98	41	1.92	20	Pass
NVLT	BLE 1M	2480	Ant1	-0.85	41	2.05	20	Pass
NVHT	BLE 1M	2402	Ant1	-1.70	41	1.20	20	Pass
NVHT	BLE 1M	2440	Ant1	-1.00	41	1.90	20	Pass
NVHT	BLE 1M	2480	Ant1	-0.86	41	2.04	20	Pass
NVNT	BLE 2M	2402	Ant1	-1.80	54	1.10	20	Pass
NVNT	BLE 2M	2440	Ant1	-1.11	53	1.79	20	Pass
NVNT	BLE 2M	2480	Ant1	-0.78	54	2.12	20	Pass
NVLT	BLE 2M	2402	Ant1	-1.77	54	1.13	20	Pass
NVLT	BLE 2M	2440	Ant1	-1.11	54	1.79	20	Pass
NVLT	BLE 2M	2480	Ant1	-0.78	54	2.12	20	Pass
NVHT	BLE 2M	2402	Ant1	-1.76	54	1.14	20	Pass
NVHT	BLE 2M	2440	Ant1	-1.13	54	1.77	20	Pass
NVHT	BLE 2M	2480	Ant1	-0.79	54	2.11	20	Pass

Test Graphs

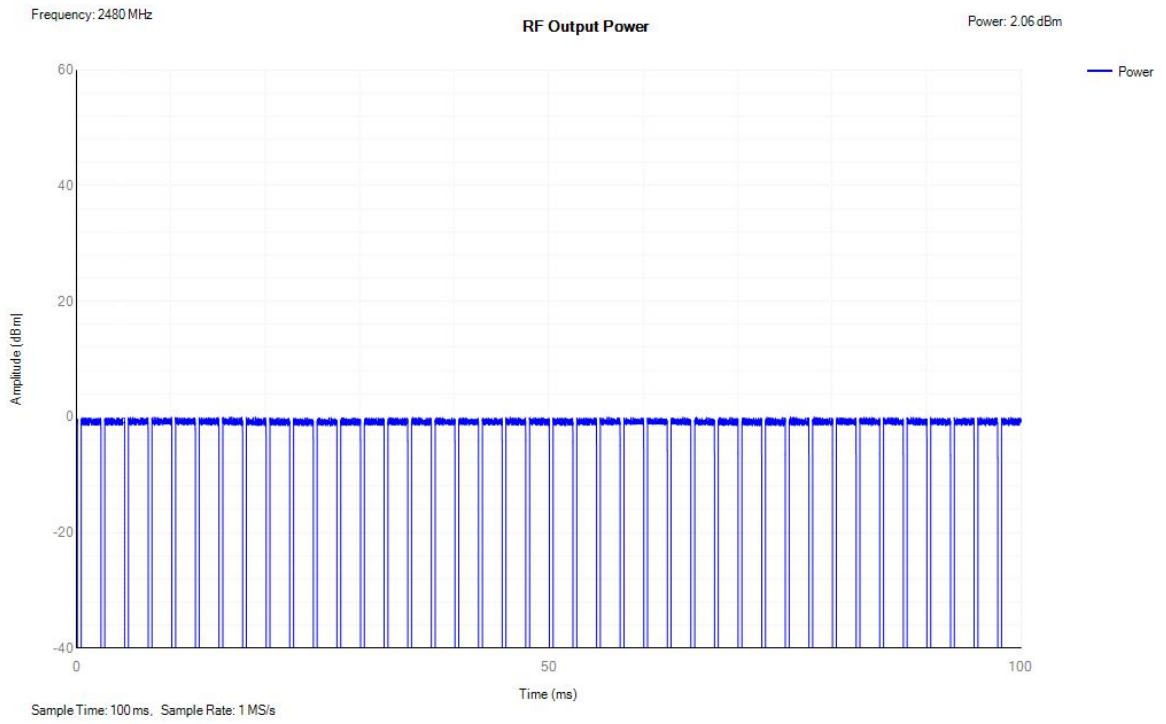
Power NVNT BLE 1M 2402MHz Ant1



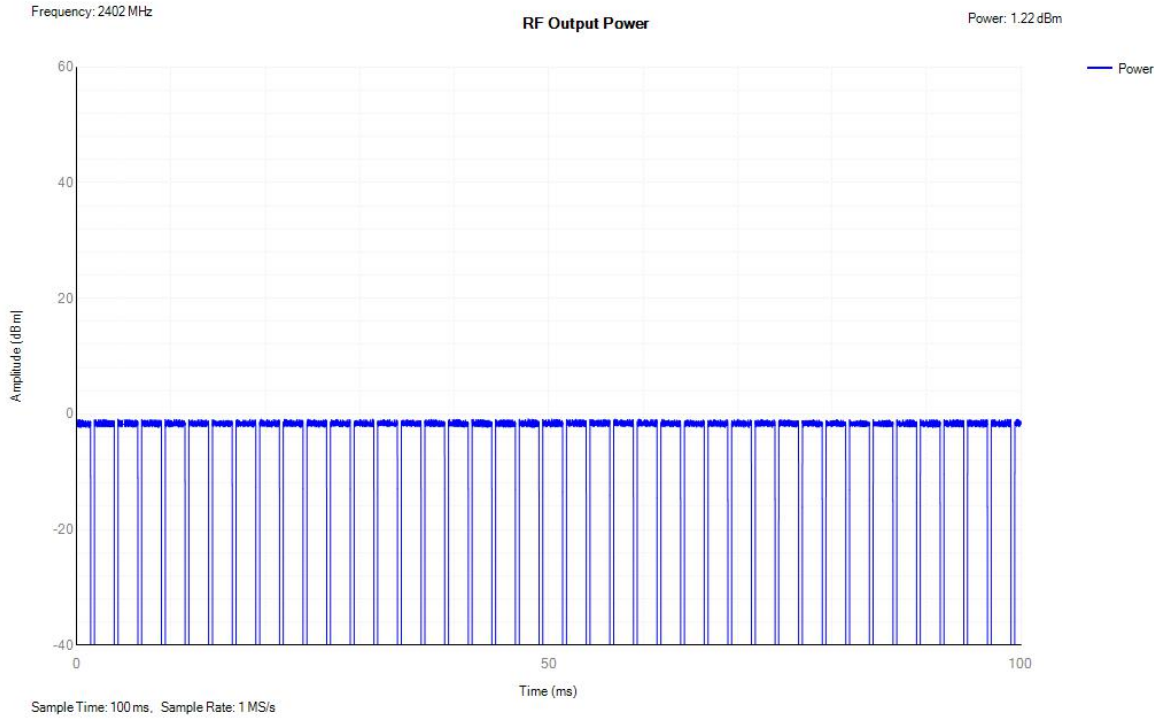
Power NVNT BLE 1M 2440MHz Ant1



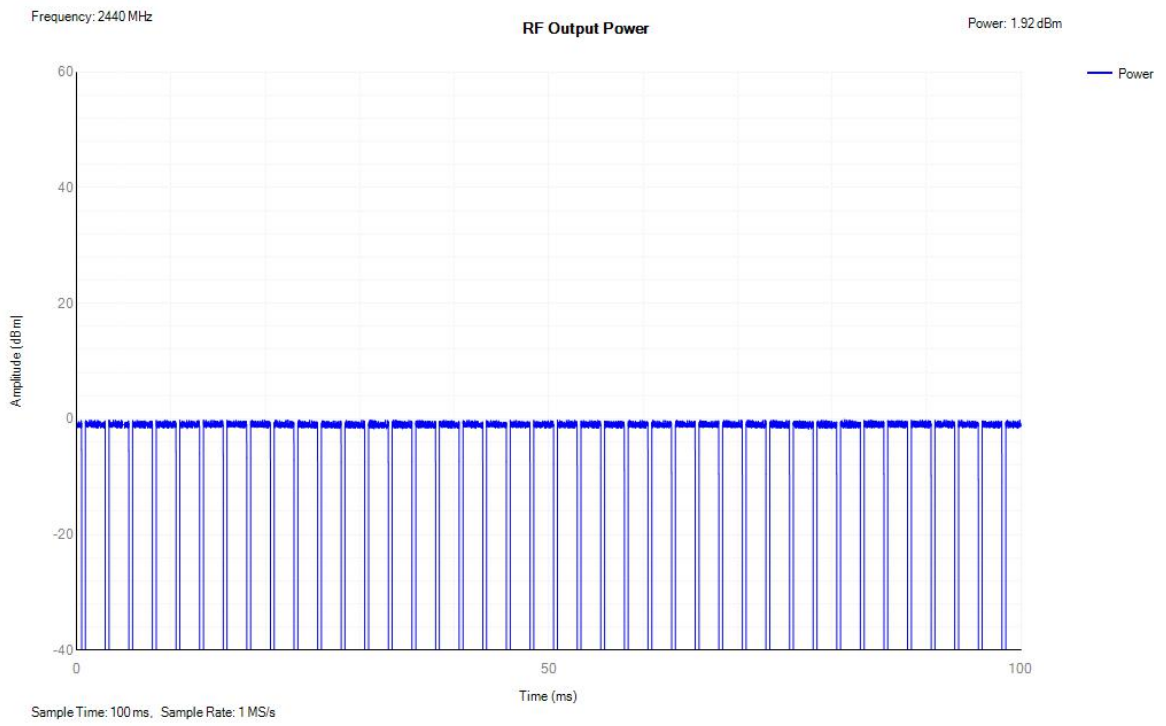
Power NVNT BLE 1M 2480MHz Ant1



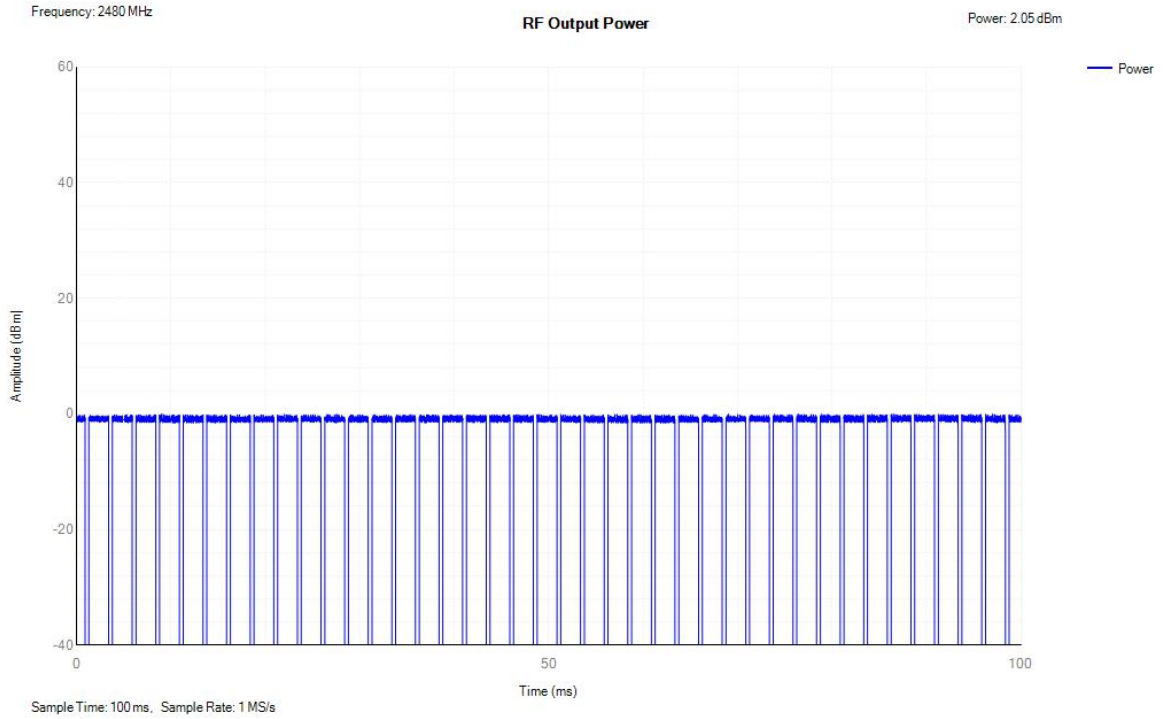
Power NVLT BLE 1M 2402MHz Ant1



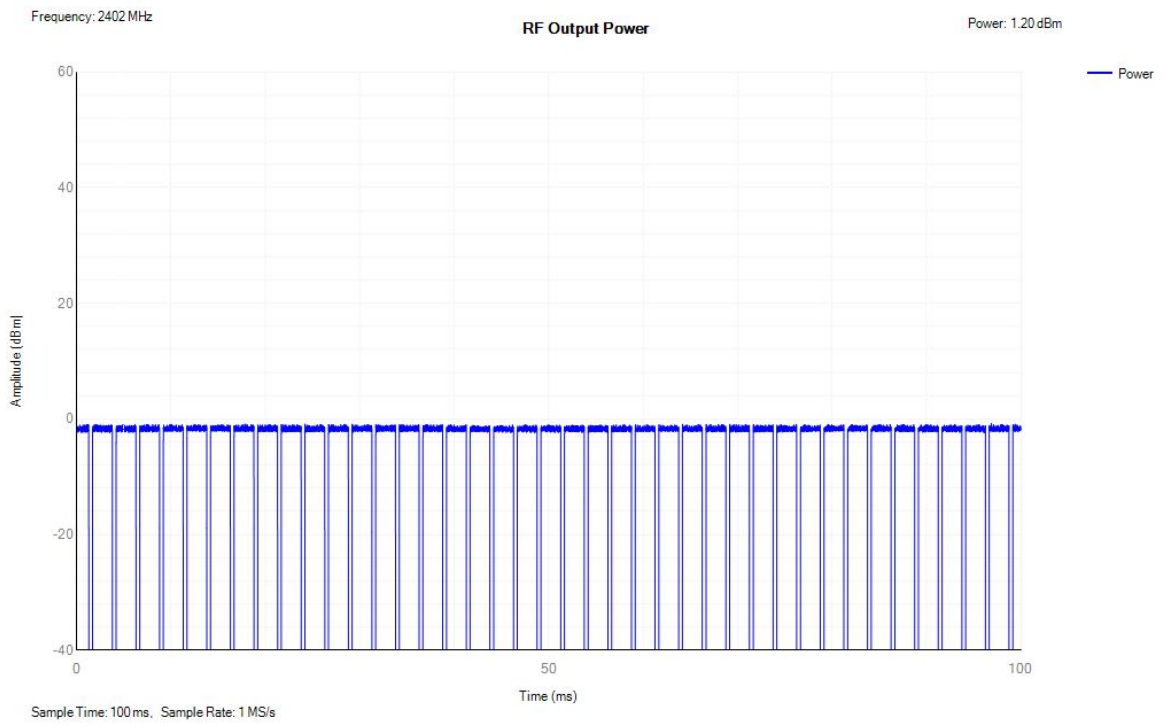
Power NVLT BLE 1M 2440MHz Ant1



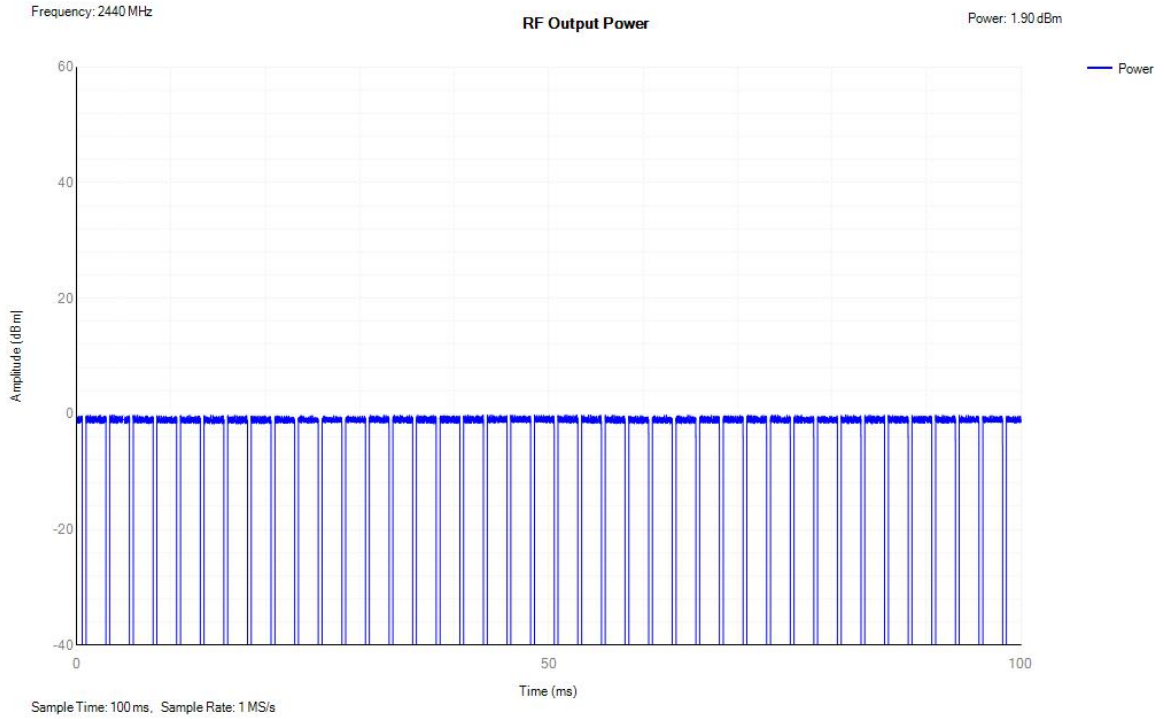
Power NVLT BLE 1M 2480MHz Ant1



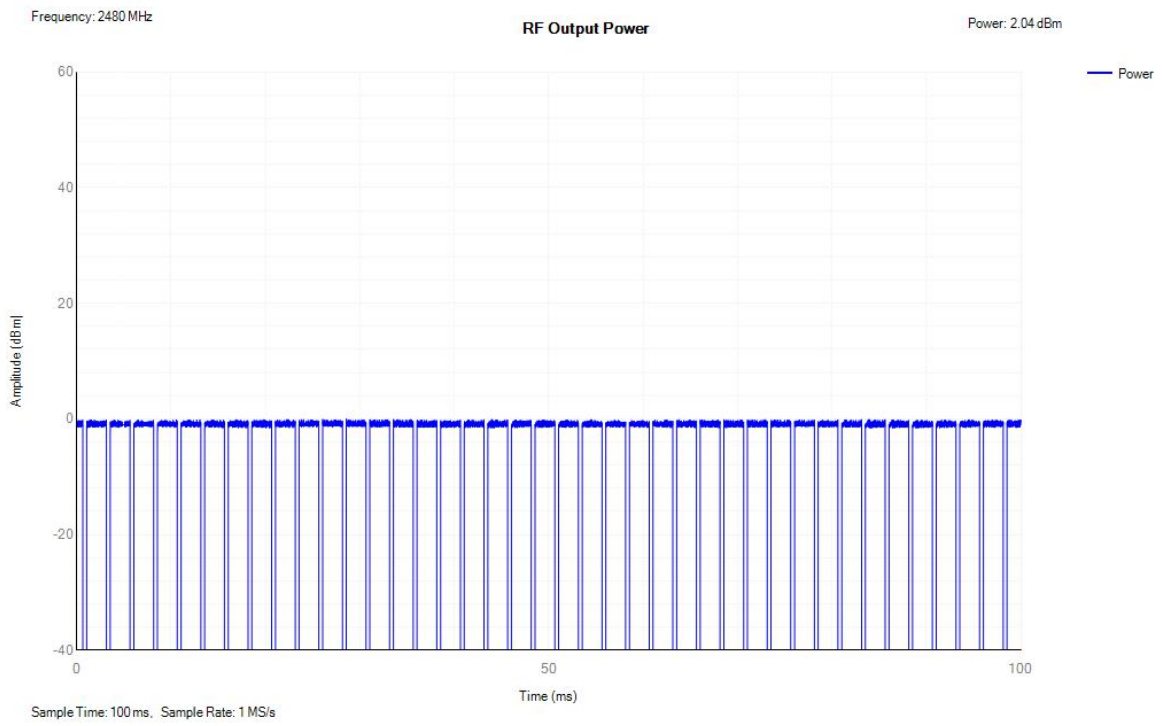
Power NVHT BLE 1M 2402MHz Ant1



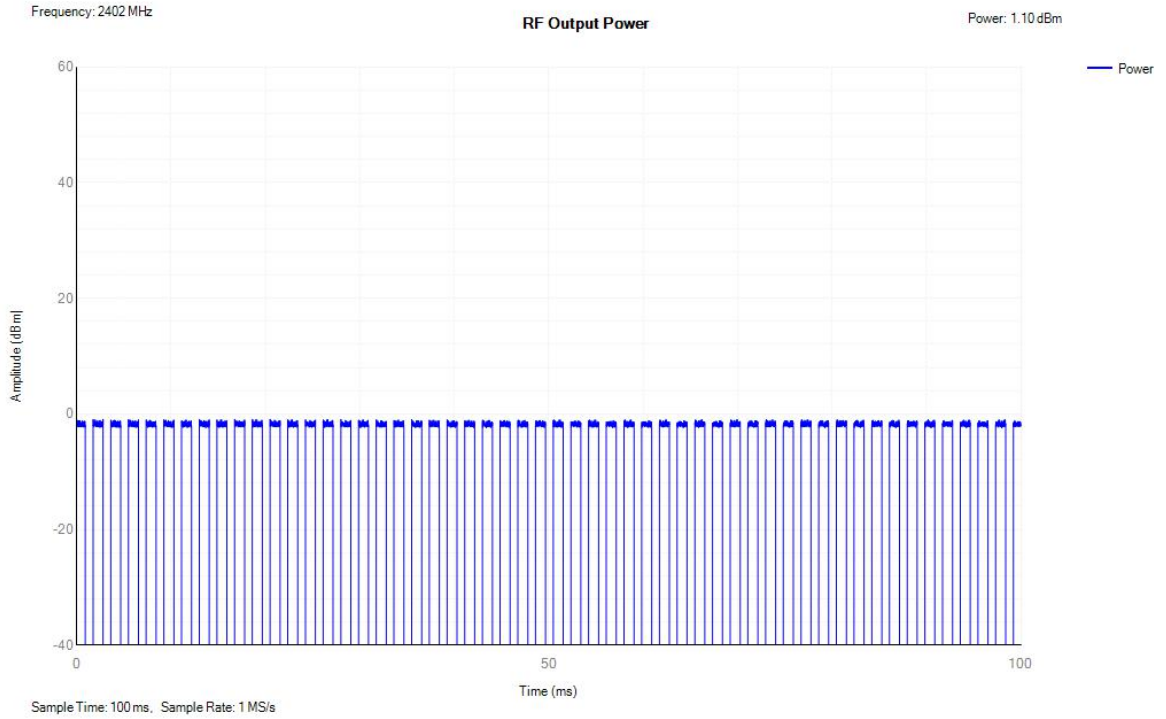
Power NVHT BLE 1M 2440MHz Ant1



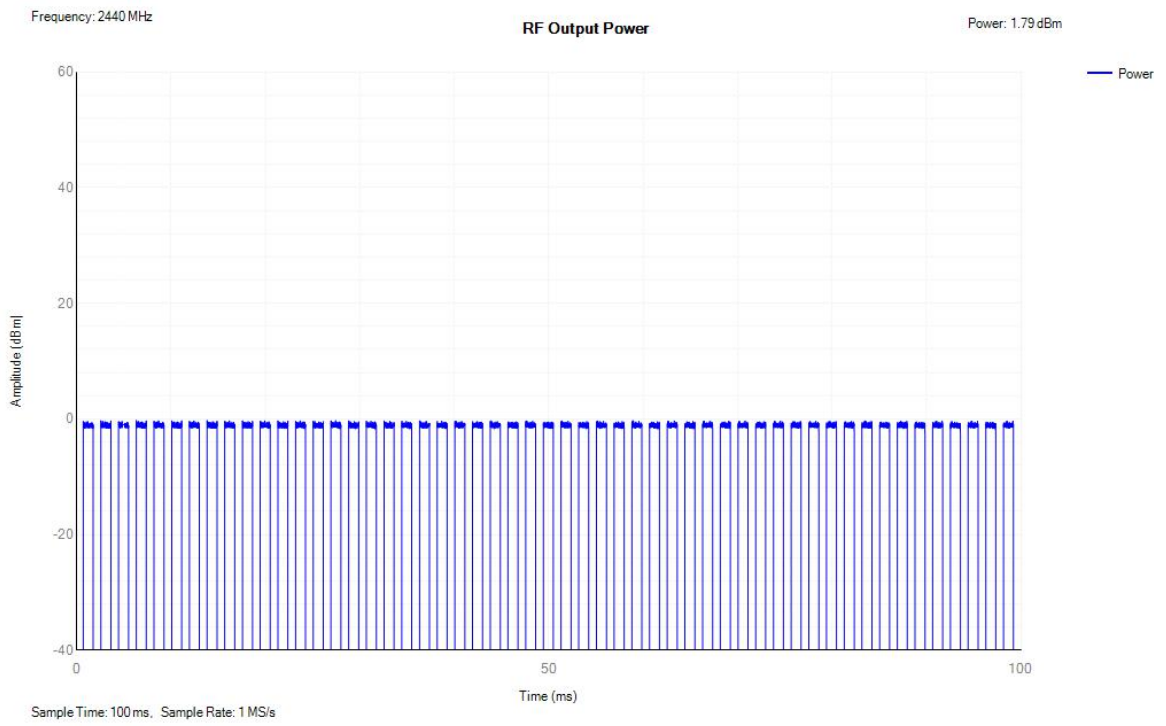
Power NVHT BLE 1M 2480MHz Ant1



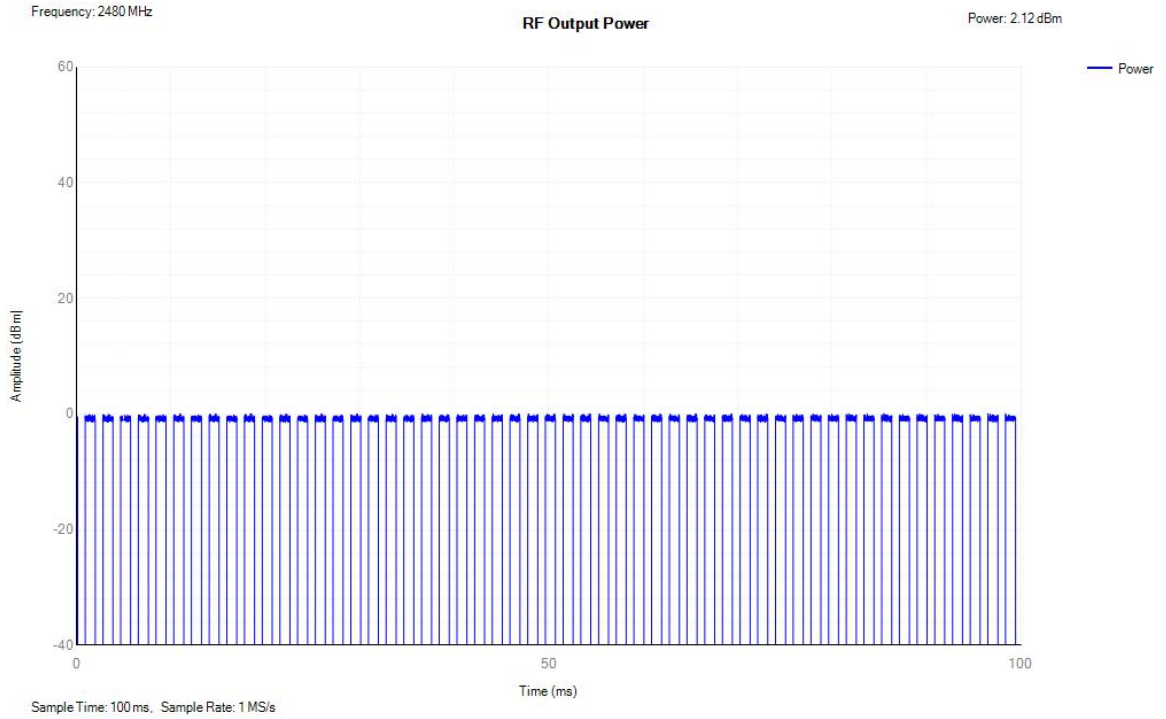
Power NVNT BLE 2M 2402MHz Ant1



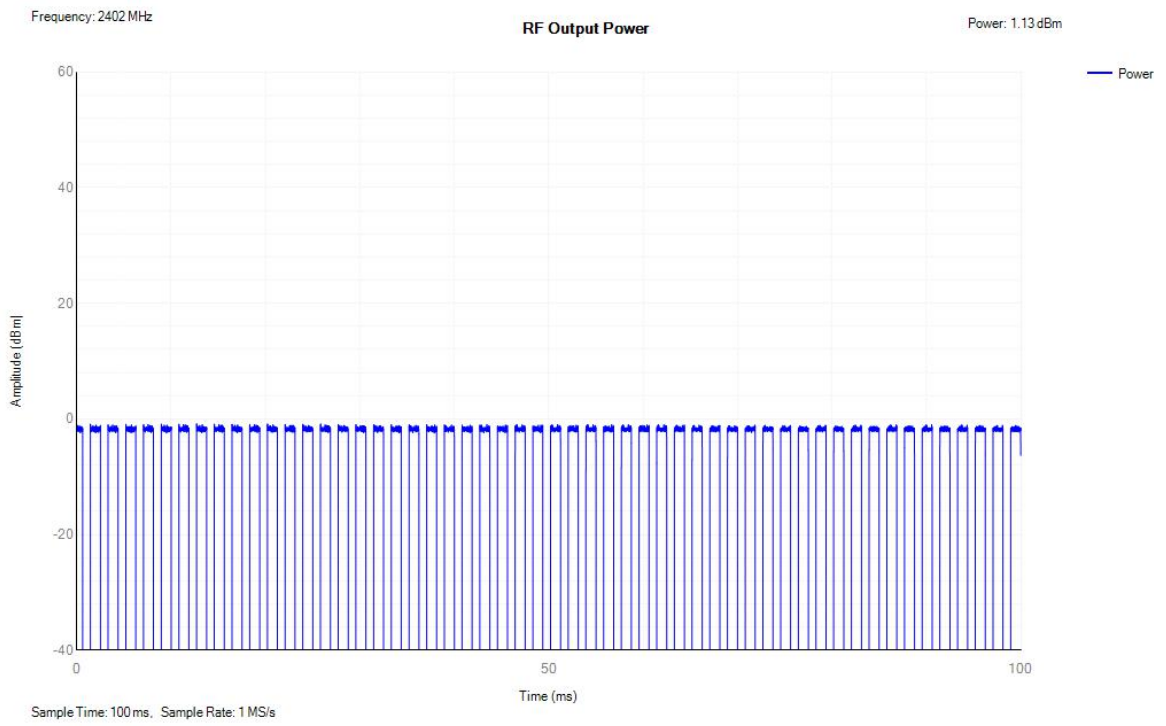
Power NVNT BLE 2M 2440MHz Ant1



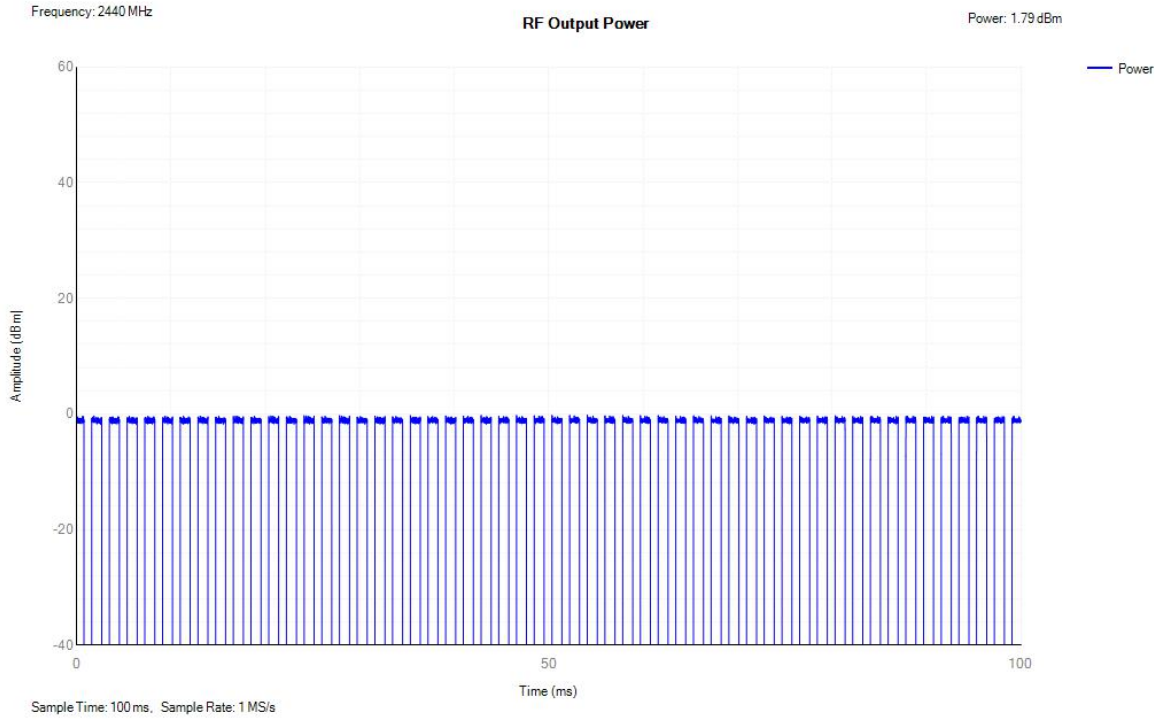
Power NVNT BLE 2M 2480MHz Ant1



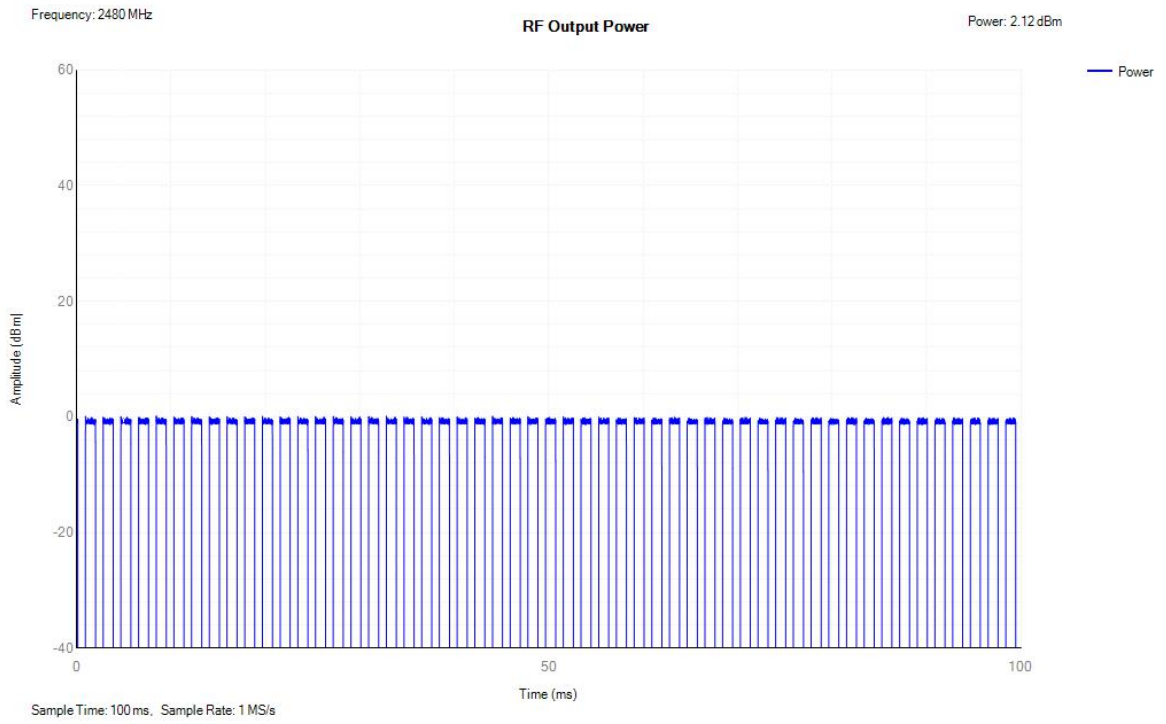
Power NVLT BLE 2M 2402MHz Ant1



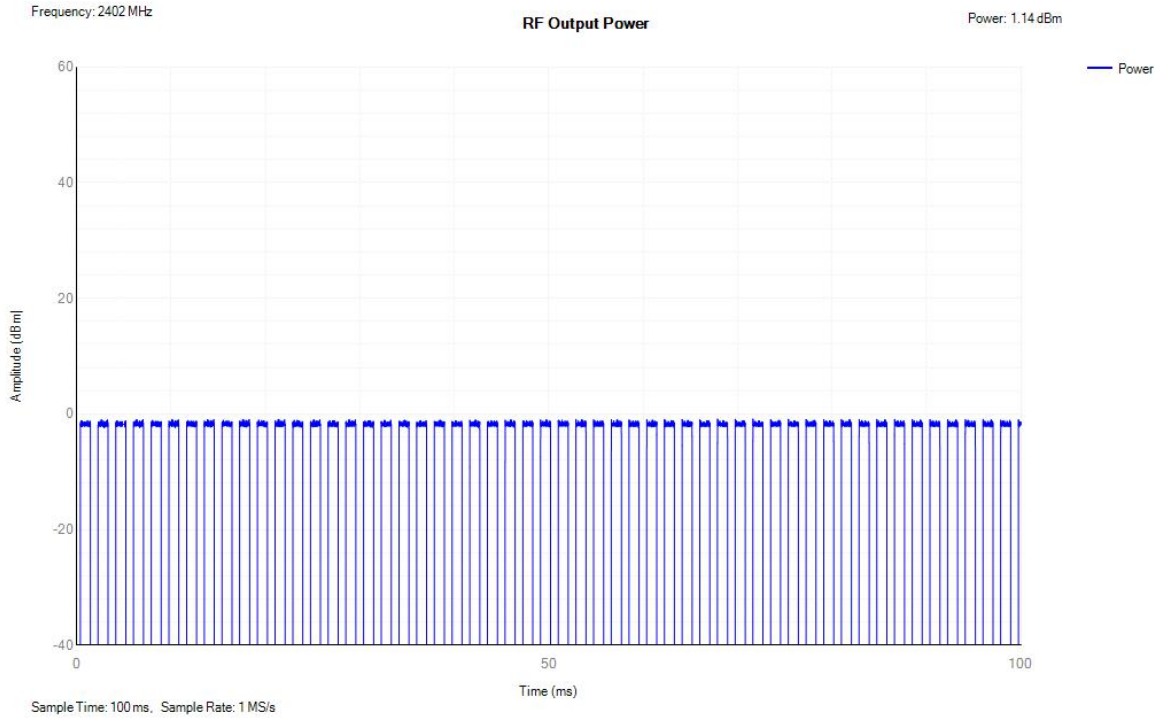
Power NVLT BLE 2M 2440MHz Ant1



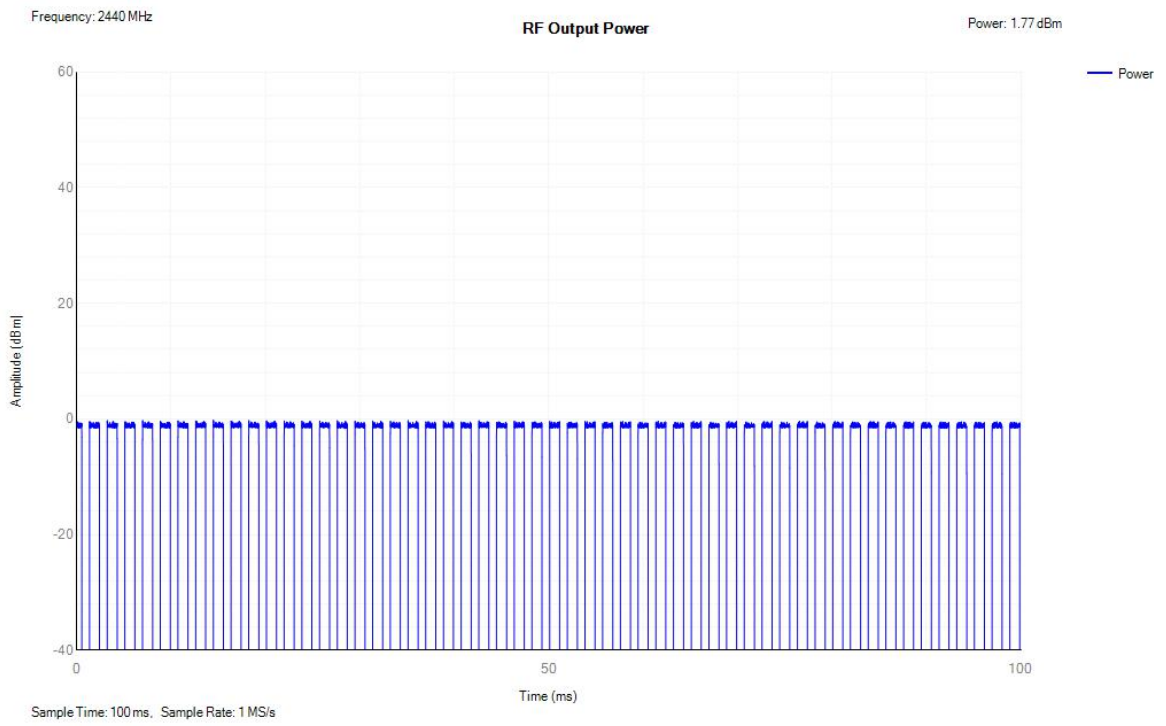
Power NVLT BLE 2M 2480MHz Ant1



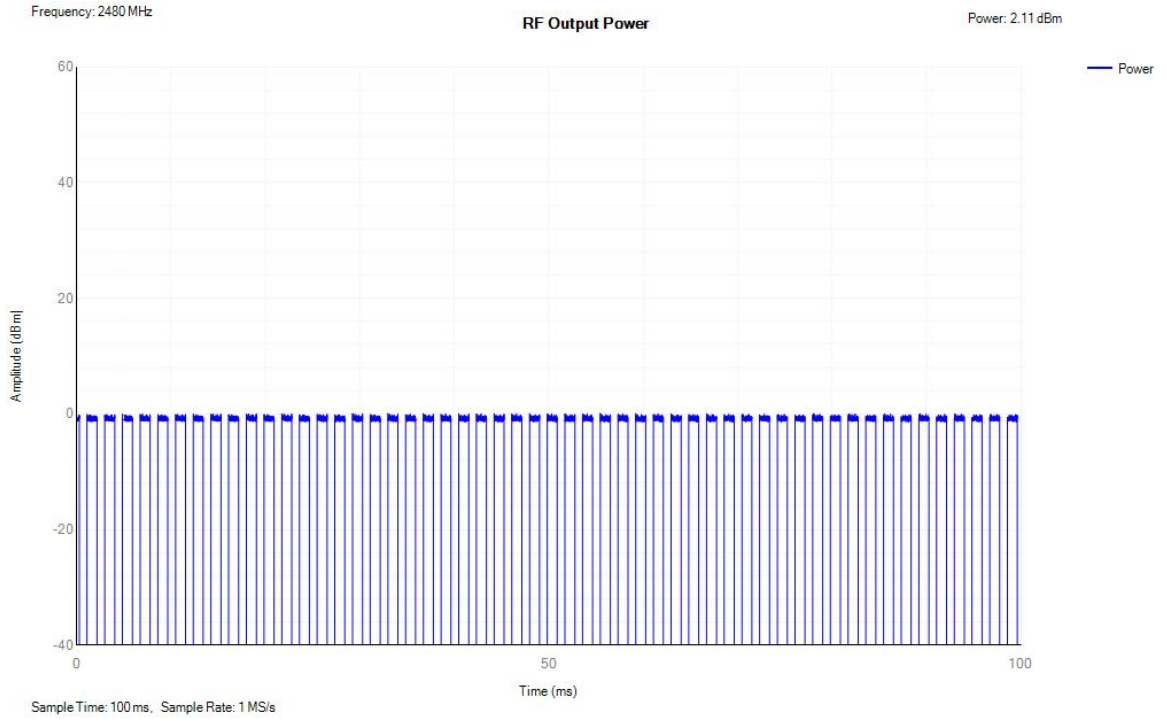
Power NVHT BLE 2M 2402MHz Ant1



Power NVHT BLE 2M 2440MHz Ant1



Power NVHT BLE 2M 2480MHz Ant1

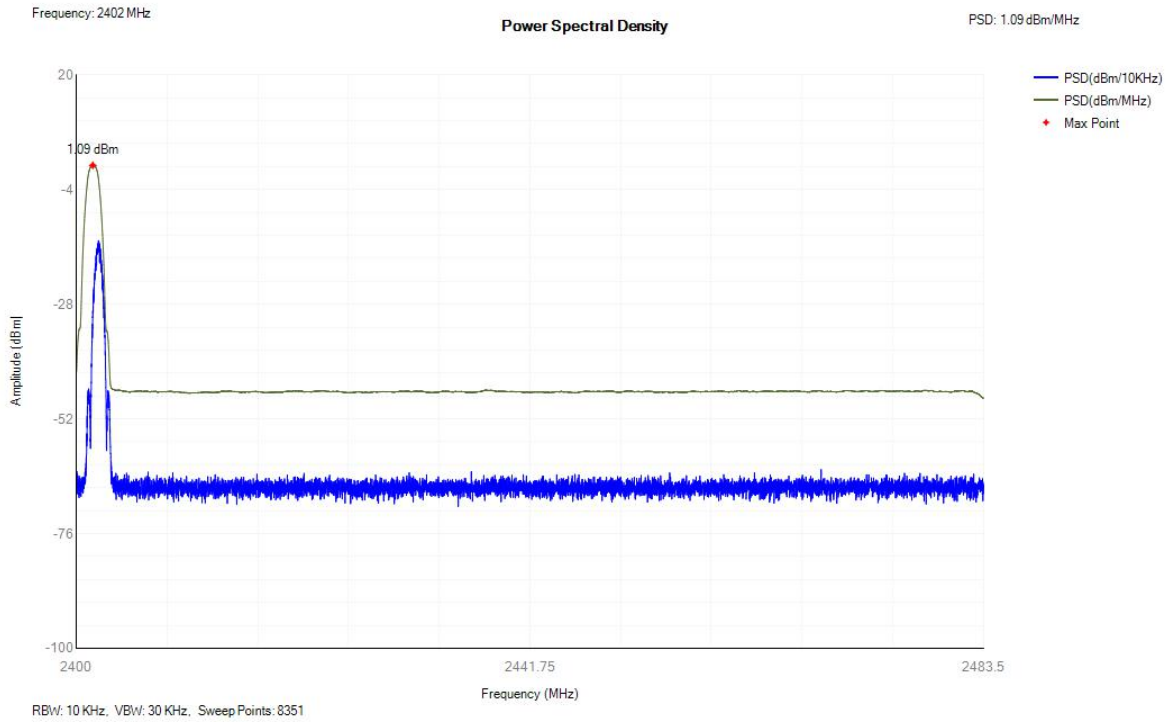


Power Spectral Density

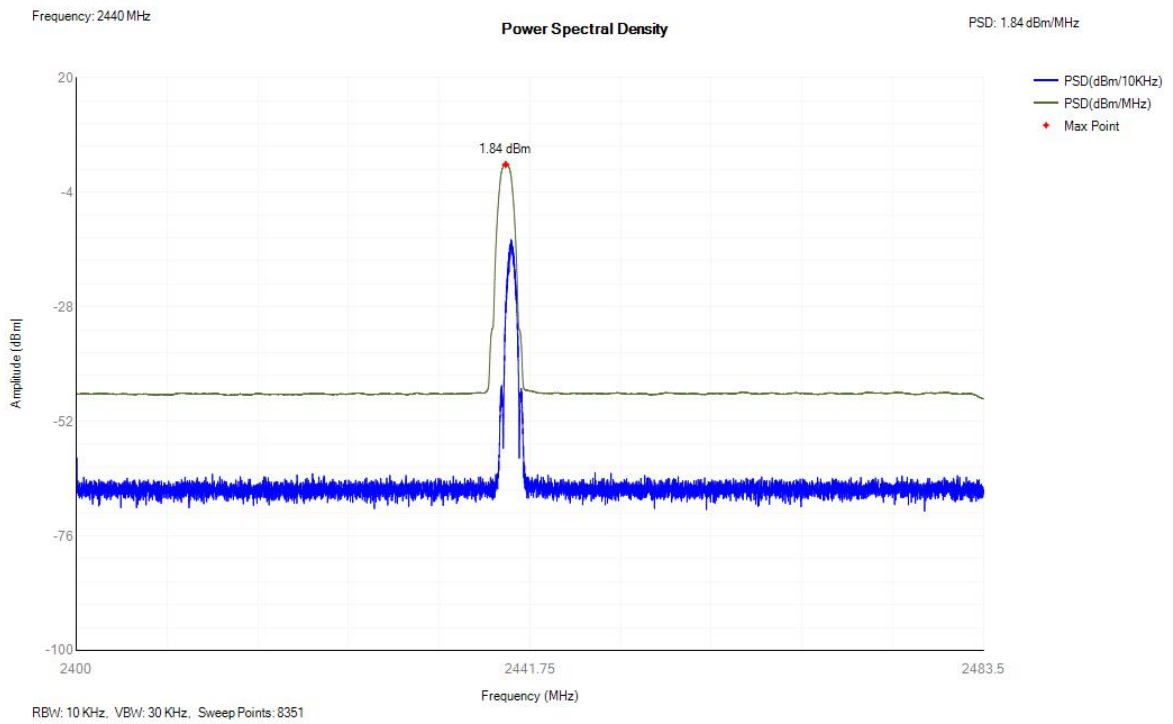
Condition	Mode	Frequency (MHz)	Antenna	Max PSD (dBm/MHz)	Limit (dBm/MHz)	Verdict
NVNT	BLE 1M	2402	Ant1	1.09	10	Pass
NVNT	BLE 1M	2440	Ant1	1.84	10	Pass
NVNT	BLE 1M	2480	Ant1	1.98	10	Pass
NVNT	BLE 2M	2402	Ant1	0.09	10	Pass
NVNT	BLE 2M	2440	Ant1	0.72	10	Pass
NVNT	BLE 2M	2480	Ant1	1.06	10	Pass

Test Graphs

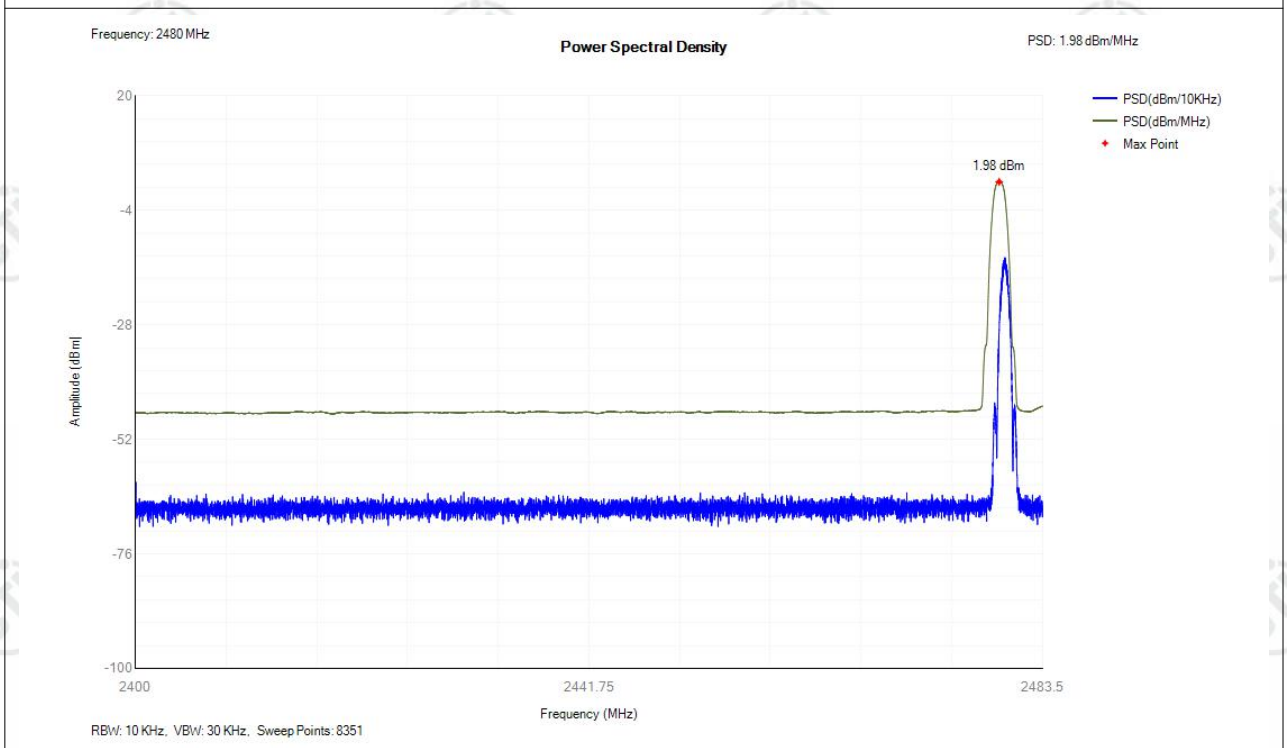
PSD NVNT BLE 1M 2402MHz Ant1



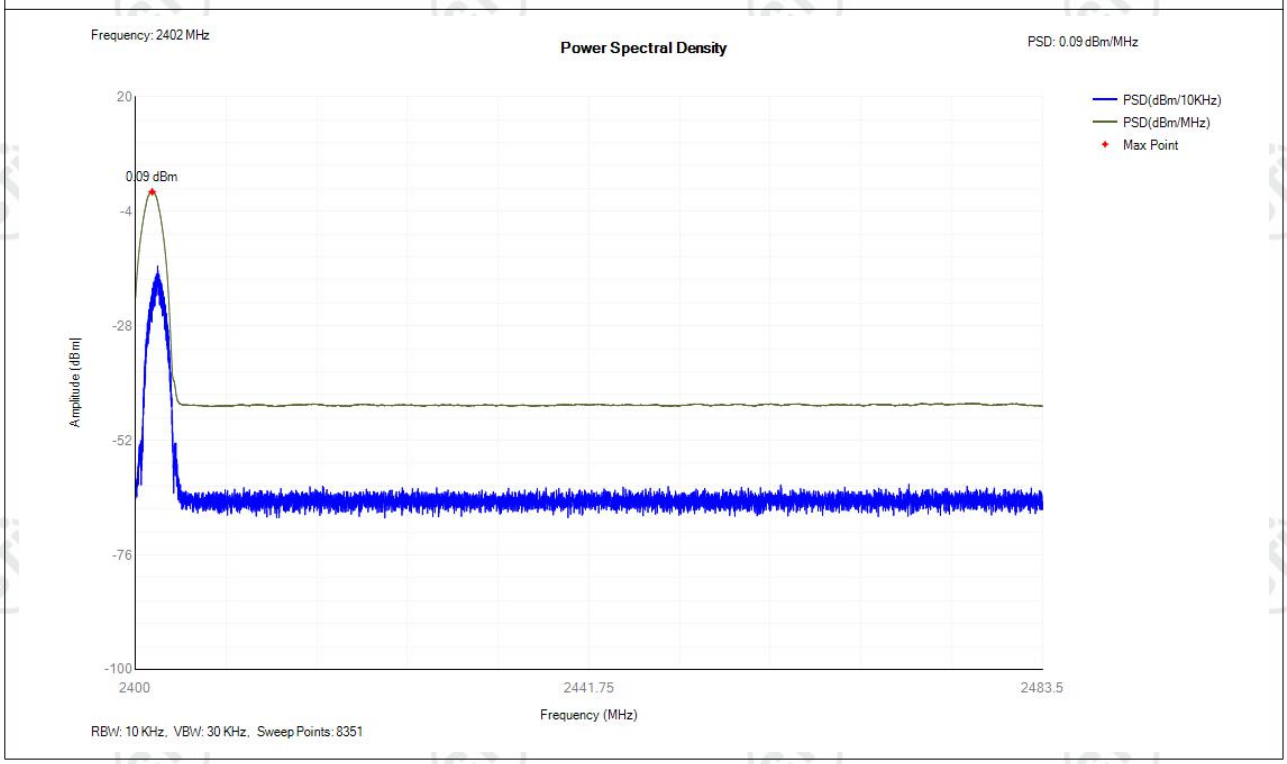
PSD NVNT BLE 1M 2440MHz Ant1



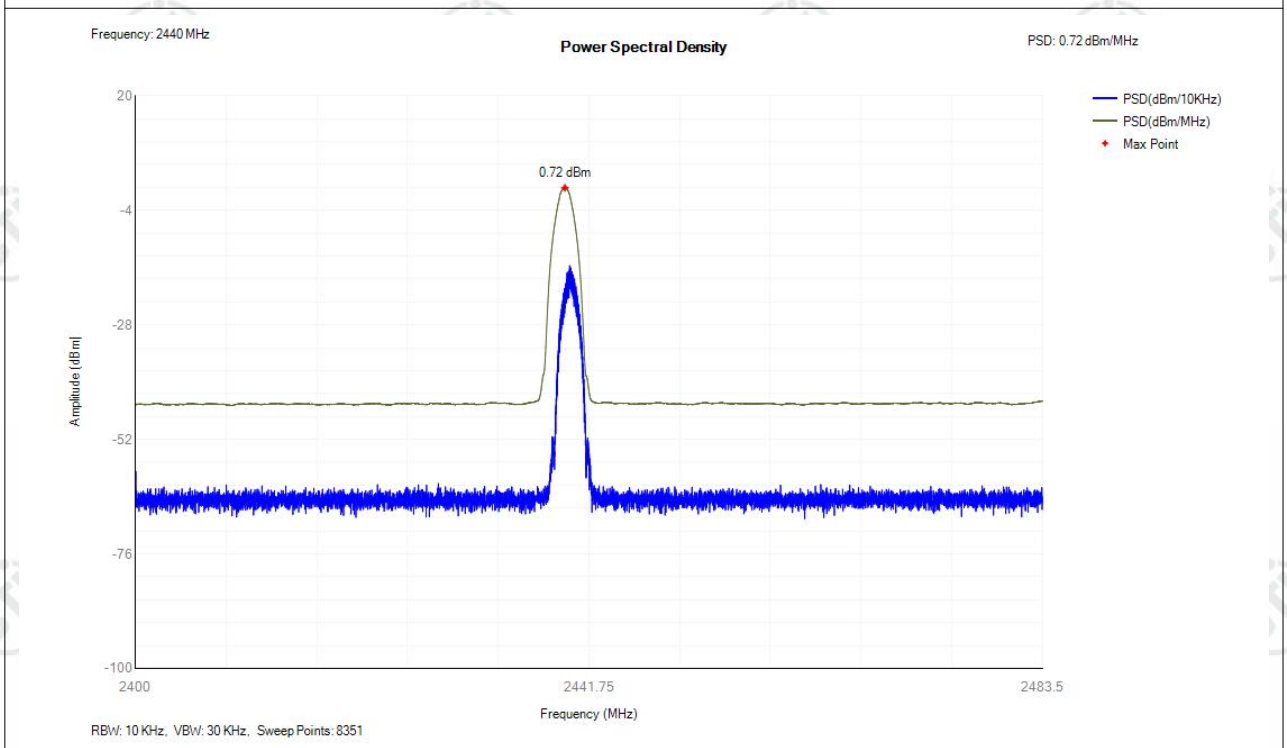
PSD NVNT BLE 1M 2480MHz Ant1



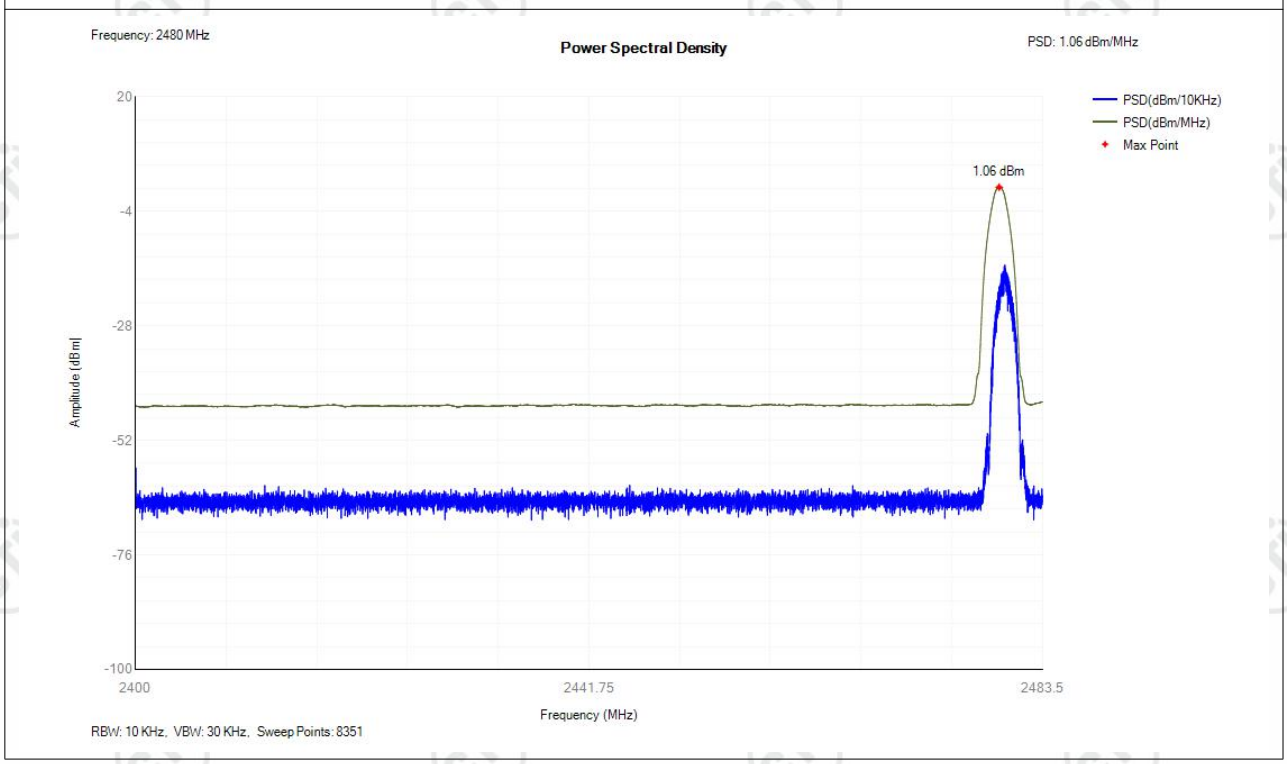
PSD NVNT BLE 2M 2402MHz Ant1



PSD NVNT BLE 2M 2440MHz Ant1



PSD NVNT BLE 2M 2480MHz Ant1

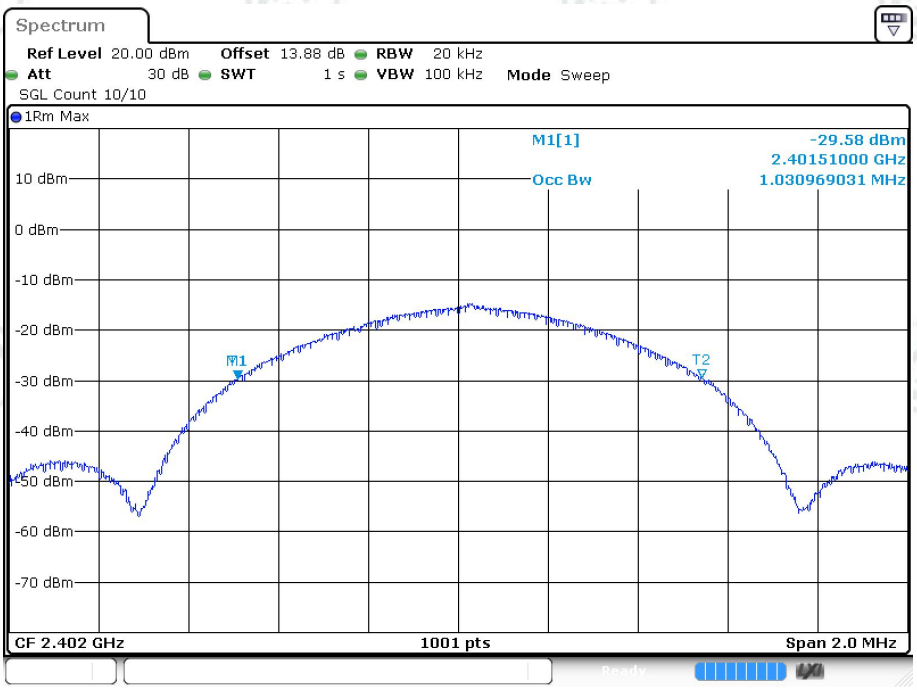


Occupied Channel Bandwidth

Condition	Mode	Frequency (MHz)	Antenna	Center Frequency (MHz)	OBW (MHz)	Lower Edge (MHz)	Upper Edge (MHz)	Limit OBW (MHz)	Verdict
NVNT	BLE 1M	2402	Ant1	2402.026	1.031	2401.51	2402.541	2400 - 2483.5MHz	Pass
NVNT	BLE 1M	2440	Ant1	2440.026	1.031	2439.51	2440.541	2400 - 2483.5MHz	Pass
NVNT	BLE 1M	2480	Ant1	2480.027	1.033	2479.51	2480.543	2400 - 2483.5MHz	Pass
NVNT	BLE 2M	2402	Ant1	2402.024	2.046	2401.001	2403.047	2400 - 2483.5MHz	Pass
NVNT	BLE 2M	2440	Ant1	2440.024	2.046	2439.001	2441.047	2400 - 2483.5MHz	Pass
NVNT	BLE 2M	2480	Ant1	2480.024	2.046	2479.001	2481.047	2400 - 2483.5MHz	Pass

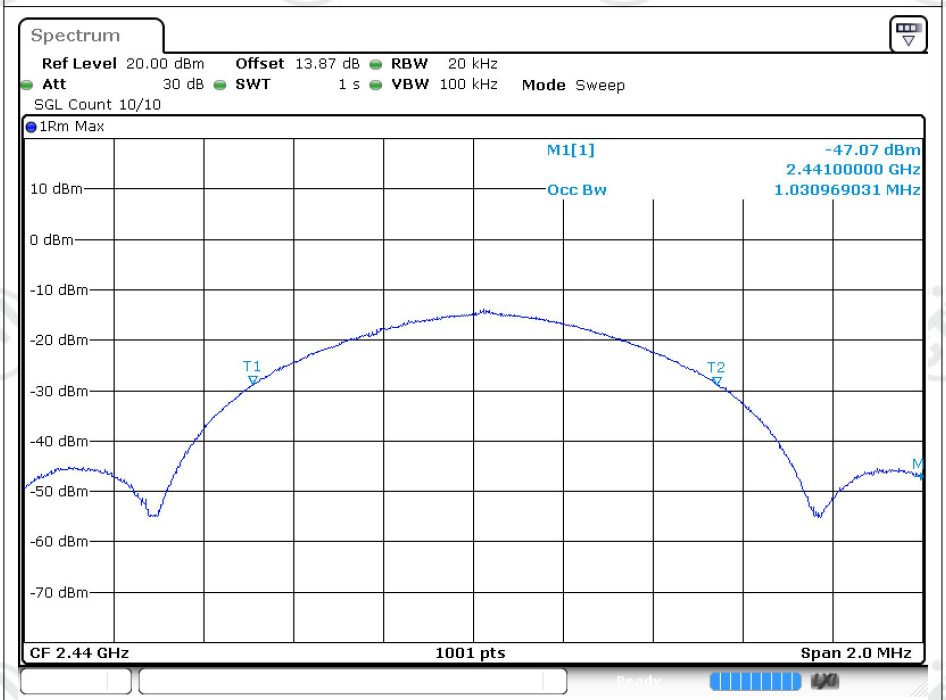
Test Graphs

OBW NVNT BLE 1M 2402MHz Ant1



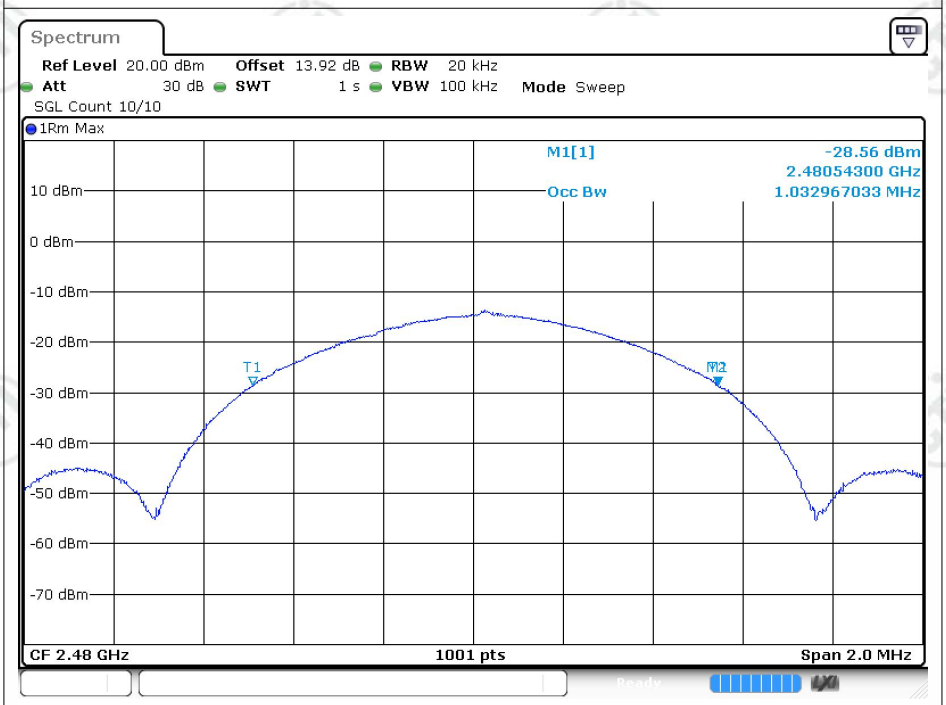
Date: 4.APR.2023 09:21:20

OBW NVNT BLE 1M 2440MHz Ant1



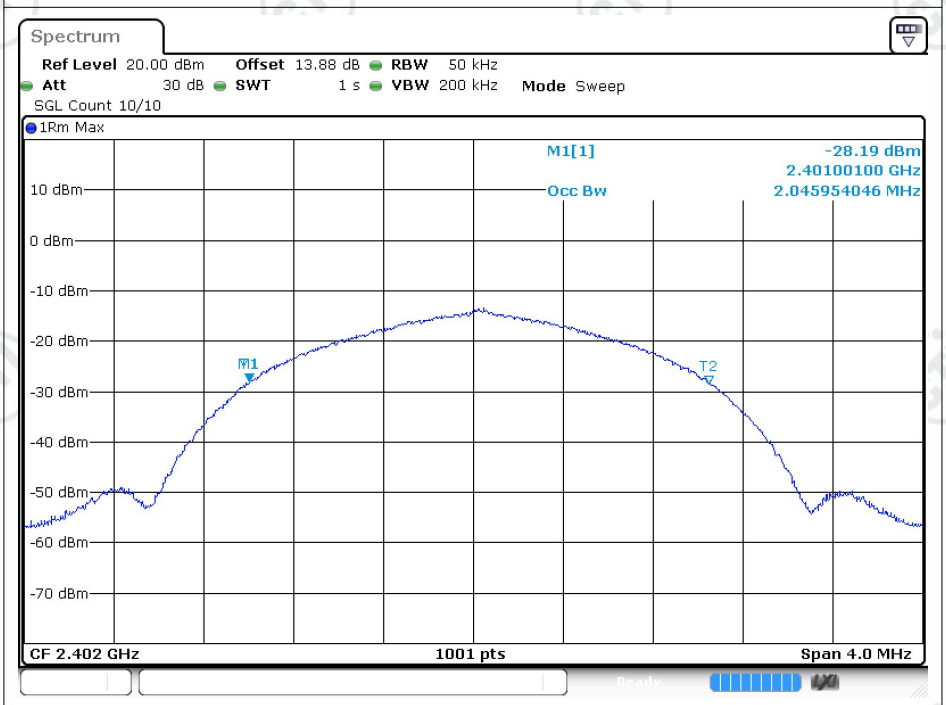
Date: 4.APR.2023 09:20:31

OBW NVNT BLE 1M 2480MHz Ant1



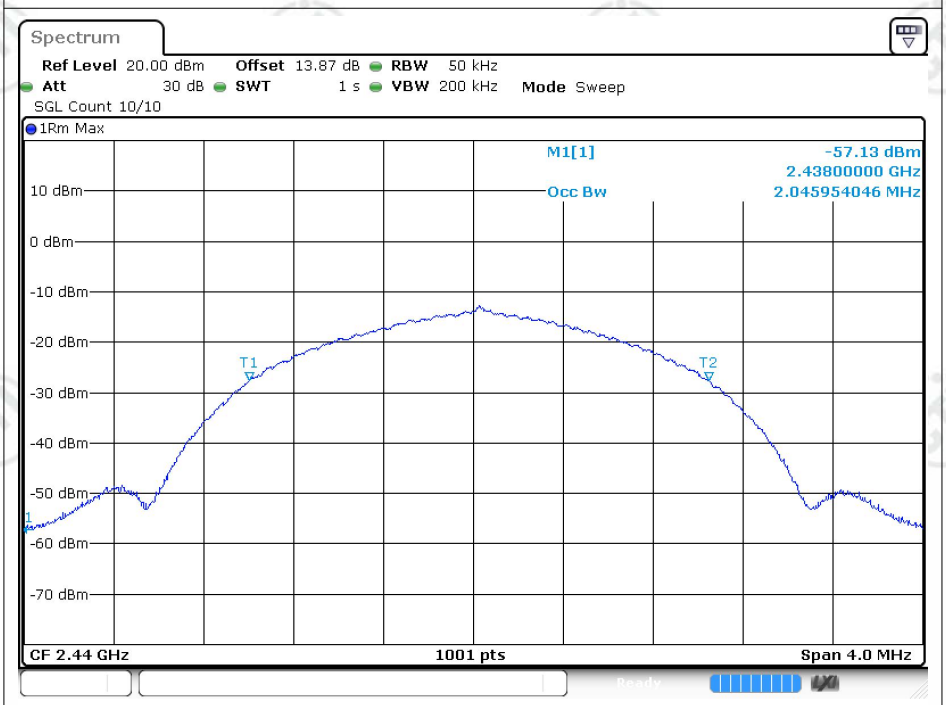
Date: 4.APR.2023 09:24:16

OBW NVNT BLE 2M 2402MHz Ant1



Date: 4.APR.2023 09:26:13

OBW NVNT BLE 2M 2440MHz Ant1



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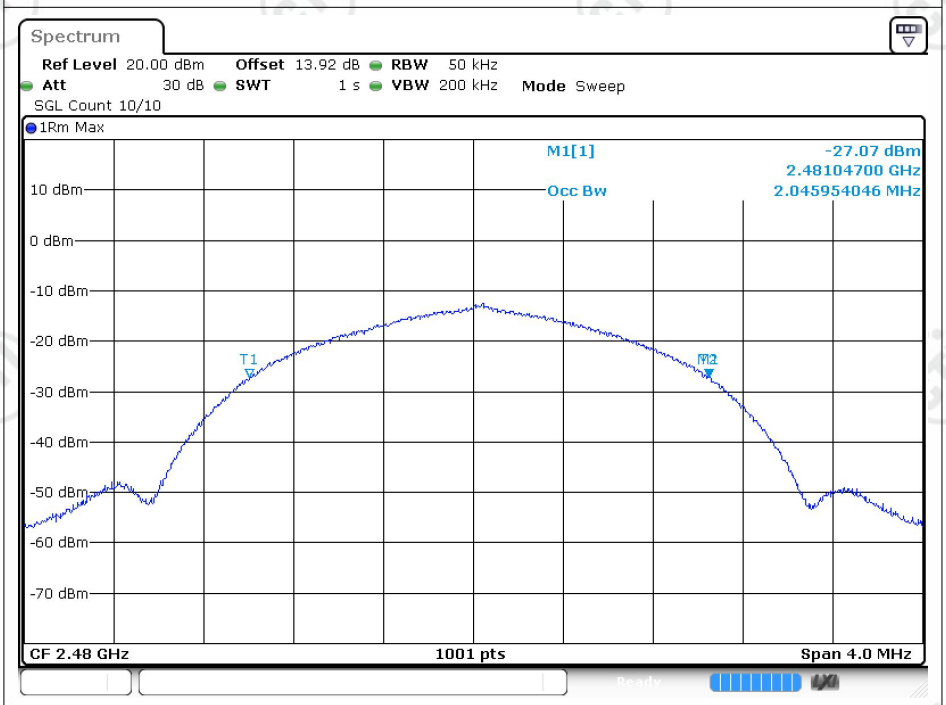
OBW NVNT BLE 2M 2480MHz Ant1



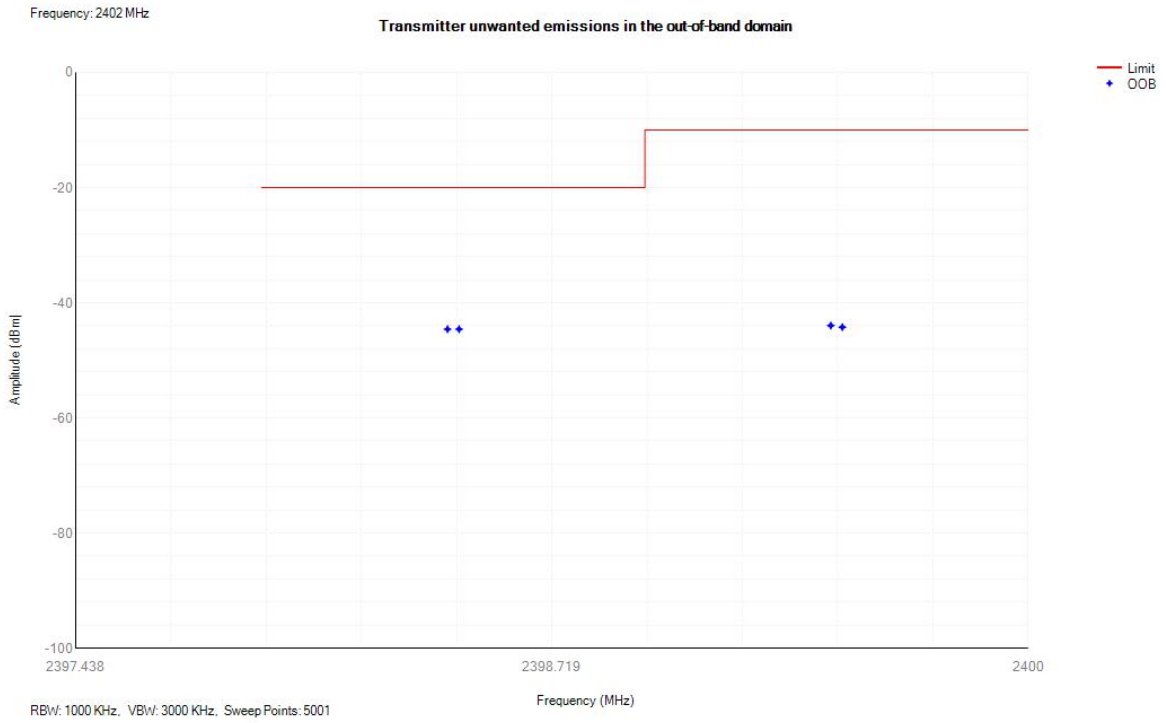
Date: 4.APR.2023 09:29:10

Transmitter unwanted emissions in the out-of-band domain

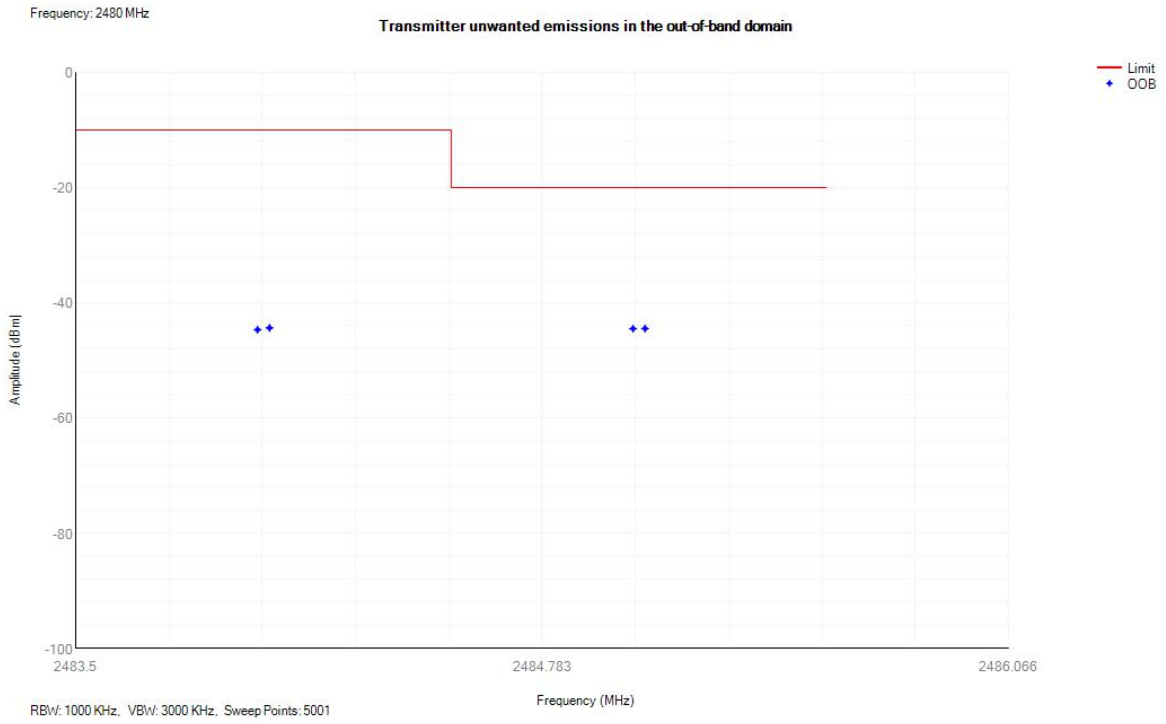
Condition	Mode	Frequency (MHz)	Antenna	OOB Frequency (MHz)	Level (dBm/MHz)	Limit (dBm/MHz)	Verdict
NVNT	BLE 1M	2402	Ant1	2399.5	-44.22	-10	Pass
NVNT	BLE 1M	2402	Ant1	2399.469	-43.95	-10	Pass
NVNT	BLE 1M	2402	Ant1	2398.469	-44.57	-20	Pass
NVNT	BLE 1M	2402	Ant1	2398.438	-44.58	-20	Pass
NVNT	BLE 1M	2480	Ant1	2484	-44.67	-10	Pass
NVNT	BLE 1M	2480	Ant1	2484.033	-44.36	-10	Pass
NVNT	BLE 1M	2480	Ant1	2485.033	-44.48	-20	Pass
NVNT	BLE 1M	2480	Ant1	2485.066	-44.46	-20	Pass
NVNT	BLE 2M	2402	Ant1	2399.5	-42.67	-10	Pass
NVNT	BLE 2M	2402	Ant1	2398.5	-43.98	-10	Pass
NVNT	BLE 2M	2402	Ant1	2398.454	-44.04	-10	Pass
NVNT	BLE 2M	2402	Ant1	2397.454	-44.28	-20	Pass
NVNT	BLE 2M	2402	Ant1	2396.454	-44.35	-20	Pass
NVNT	BLE 2M	2402	Ant1	2396.408	-44.34	-20	Pass
NVNT	BLE 2M	2480	Ant1	2484	-44.11	-10	Pass
NVNT	BLE 2M	2480	Ant1	2485	-44.01	-10	Pass
NVNT	BLE 2M	2480	Ant1	2485.046	-44.05	-10	Pass
NVNT	BLE 2M	2480	Ant1	2486.046	-44.10	-20	Pass
NVNT	BLE 2M	2480	Ant1	2487.046	-44.19	-20	Pass
NVNT	BLE 2M	2480	Ant1	2487.092	-44.13	-20	Pass

Test Graphs

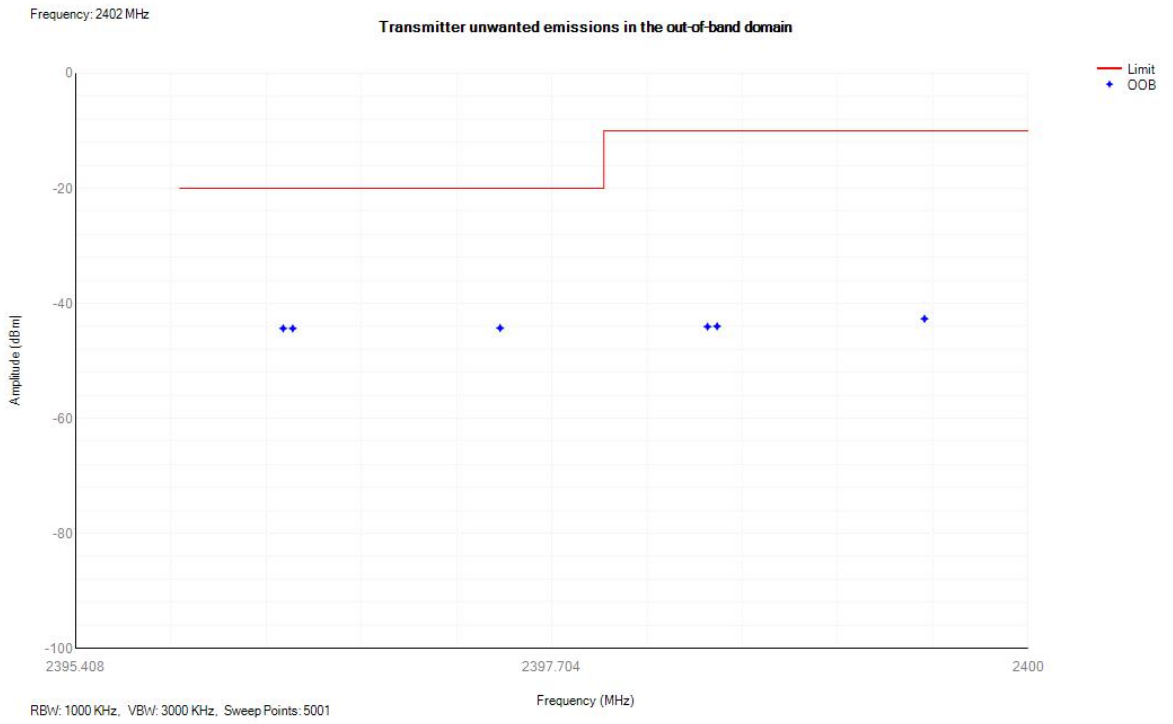
Tx. Emissions OOB NVNT BLE 1M 2402MHz Ant1



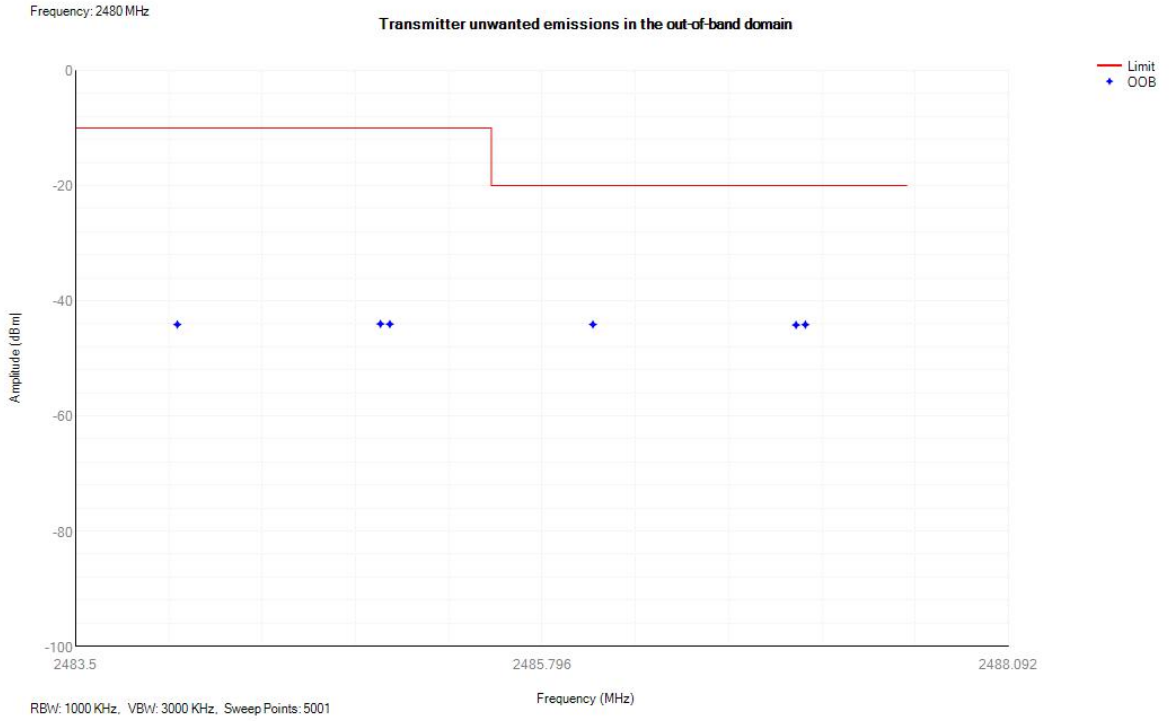
Tx. Emissions OOB NVNT BLE 1M 2480MHz Ant1



Tx. Emissions OOB NVNT BLE 2M 2402MHz Ant1



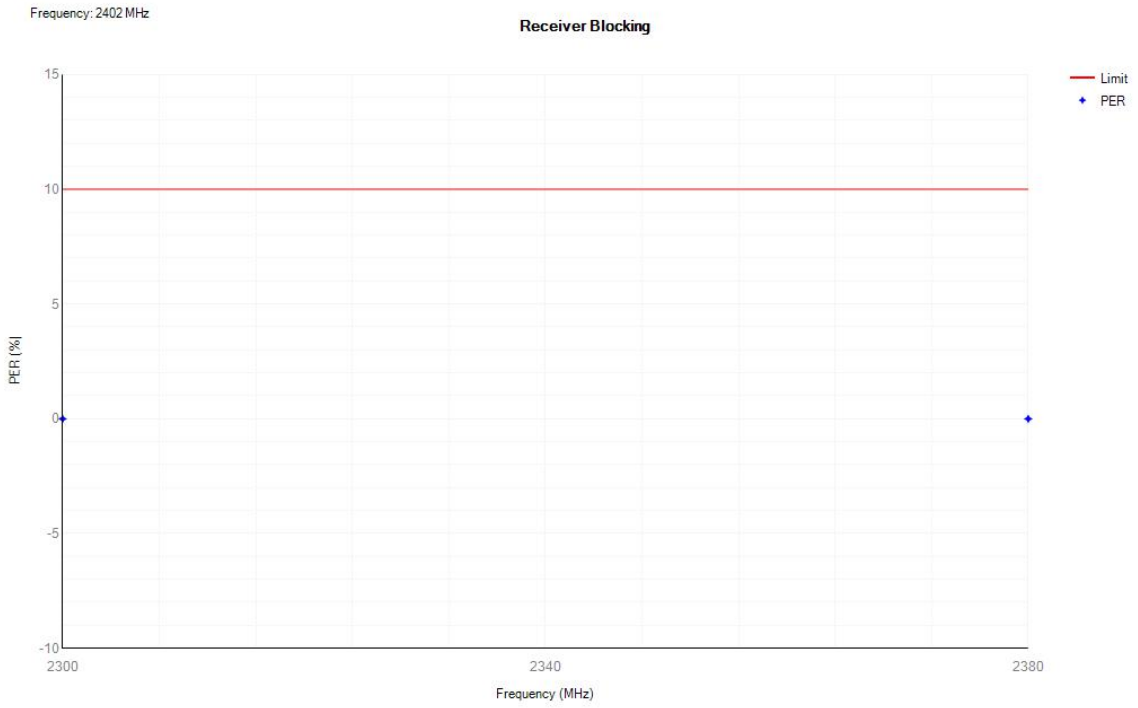
Tx. Emissions OOB NVNT BLE 2M 2480MHz Ant1



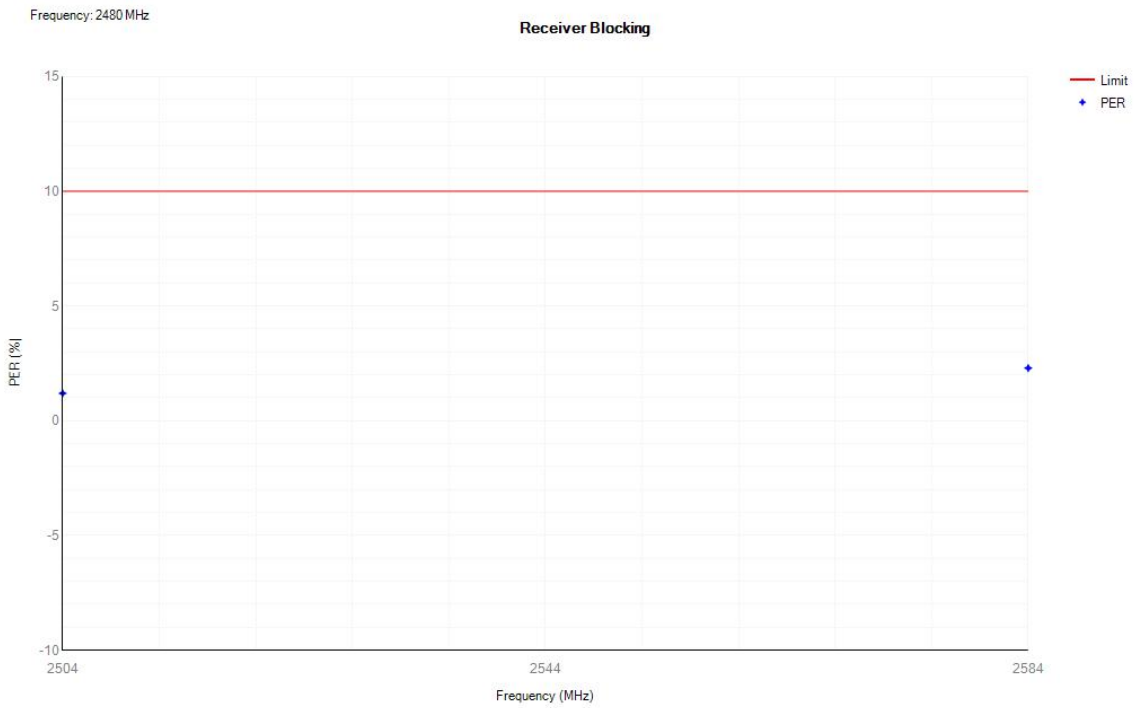
Receiver Blocking

Condition	Mode	Frequency (MHz)	Antenna	Wanted Power (dBm)	Blocking Frequency (MHz)	Blocking Power (dBm)	PER (%)	Limit (%)	Verdict
NVNT	BLE 1M	2402	Ant1	-65.97	2380	-31.1	0	10	Pass
NVNT	BLE 1M	2402	Ant1	-65.97	2300	-31.1	0	10	Pass
NVNT	BLE 1M	2480	Ant1	-65.96	2504	-31.1	1.2	10	Pass
NVNT	BLE 1M	2480	Ant1	-65.96	2584	-31.1	2.3	10	Pass

Test Graphs
Rx. Blockings NVNT BLE 1M 2402MHz Ant1



Rx. Blockings NVNT BLE 1M 2480MHz Ant1



*** End of Report ***